CORTEX PLATFORM

Technical Architecture & System Design Report

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AUTHOR: System Engineering & Architecture Group
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1. Executive Summary

The Cortex Structural Intelligence Platform is a highly optimized drone frame ingestion, facade homography stitching, defect quantification, and diagnostic reporting pipeline. Designed to assist civil engineers in identifying and tracking concrete defects (cracks, spalls, corrosion), the platform combines low-level computer vision algorithms (SIFT keypoint registration, distance transforms) with machine learning classification (ResNet-50 features + XGBoost classifier) and a robust web app architecture.

To achieve maximum throughput and avoid heavy network overhead when transferring high-resolution raw UAV flight frames (e.g. 4000x3000 px), the backend image processing core runs locally in memory sharing GPU resources, orchestrated asynchronously via Celery workers.

2. Technological Stack Configuration

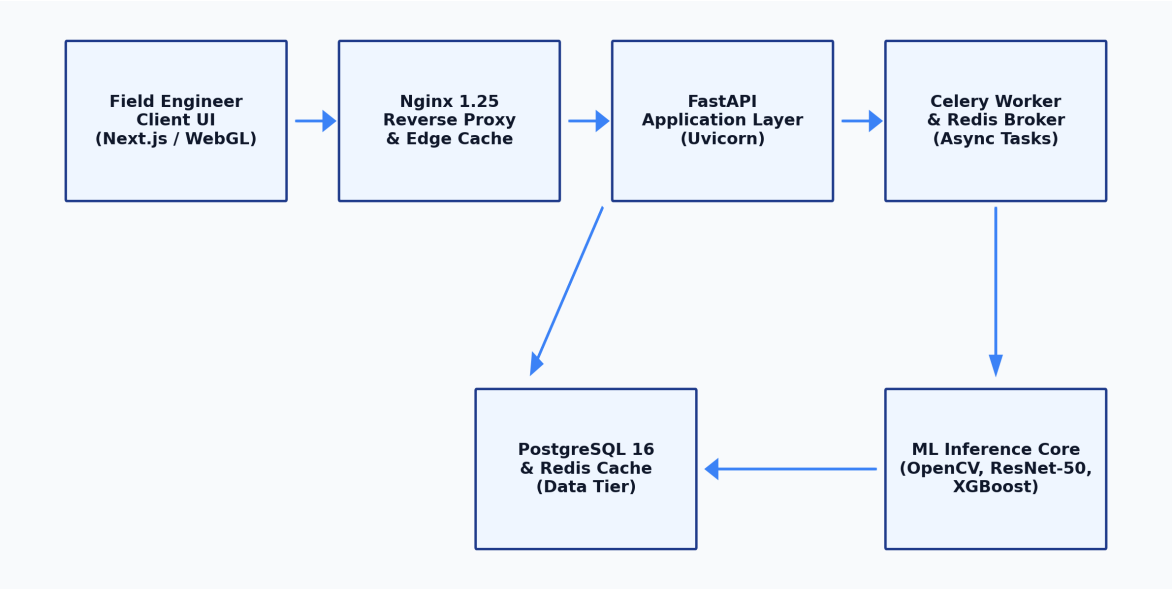
Tier	Technology	Role & Configuration
Frontend UI	Next.js 16 (React 19)	Static build served by Nginx. Dynamic custom WebGL Three.js viewport for rendering 3D structural elements (Column, Beam, Slab) with parametric crack mappings.
Reverse Proxy / Edge	Nginx 1.25	Configured as an edge cache & reverse proxy. Caches static assets, handles TLS termination, rate-limits endpoints (api_zone: 30r/s, auth_zone: 5r/m), and routes /api/* to the FastAPI backend.
Application Layer	FastAPI (Python 3.11)	Non-blocking asynchronous web layer. Implements token-bucket sliding-window rate-limiting in Redis and handles multipart/form image uploads.
Async Tasks & Queues	Celery 5.3 + Redis	Offloads resource-heavy stitching homographies, SIFT alignments, and PDF generation from the API request thread pool.
ML & Inference Core	OpenCV + ResNet-50 + XGBoost	Runs edge quality gates (Laplacian blur and exposure checks), downsampled SIFT homography registration, ResNet-50 patch feature extraction, and XGBoost false-positive filtering.
Database & Cache	PostgreSQL 16 + Redis	Relational storage for inspections, defects, and audit logs. Redis acts as a fast cache-aside storage for job state polling.

3. System Components & Architecture

Cortex utilizes a private VPC architecture. Nginx acts as the gatekeeper, routing frontend assets directly from static storage and forwarding `/api/v1` traffic to the FastAPI application servers.

3.1 Platform High-Level Architecture Flowchart

This flowchart outlines how raw drone frame inputs traverse the systems from the Field Engineer client interface through Nginx, FastAPI, Celery, and model databases:

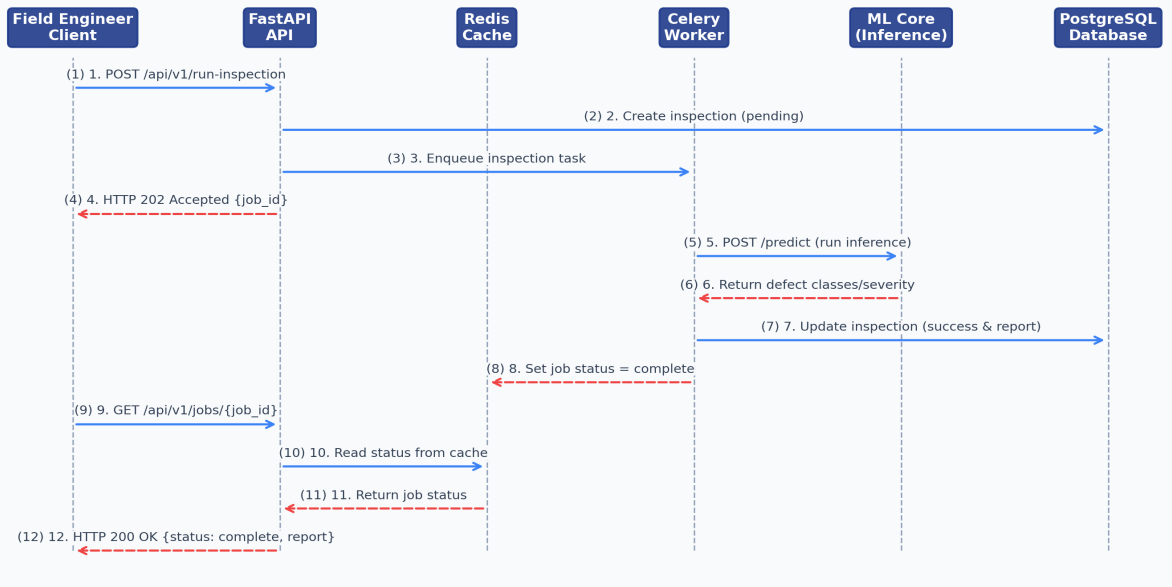


3.2 Key Architectural Flow Phases

- **Client Web Interface**: Sends UAV frames through Nginx reverse proxy.
- **FastAPI Server**: Handles rate-limiting verification, inserts a new database inspection entry in PostgreSQL with status 'PENDING', saves raw files locally, and dispatches the task asynchronously via Redis.
- **Celery Worker Queue**: Dequeues the task, performs image homography alignment, feeds crop patches to the ML layer, computes zone-severity indexes, and compiles the final PDF report.
- **Relational Database (PostgreSQL)**: Persists all inspection details and categorized defects, maintaining complete relational integrity.
- **Redis Cache**: Stores lightweight job status states (e.g. progress percentage), allowing the client to poll without database contention.

3.3 End-To-End Processing Job Lifecycle (Data Flow)

The processing lifecycle is fully asynchronous to prevent blocking user actions. Below is the sequence detailing how a facade inspection job is queued, evaluated, and compiled:



3.4 Relational Database Schema Design (DDL snippet)

```
CREATE TABLE inspections (
    id                  VARCHAR(64) PRIMARY KEY,
    building_id         VARCHAR(64) NOT NULL,
    vi_score            DOUBLE PRECISION,
    vi_class            VARCHAR(16) CHECK(vi_class IN ('minor', 'moderate', 'severe', 'critical')),
    pipeline_version    VARCHAR(16) DEFAULT '1.4.0',
    warnings            TEXT DEFAULT '[]', -- Stored as valid JSON string
    s3_key              VARCHAR(512),
    row_version         INTEGER DEFAULT 1 NOT NULL
);

CREATE TABLE defects (
    id                  SERIAL PRIMARY KEY,
    defect_id           VARCHAR(64) NOT NULL UNIQUE,
    inspection_id       VARCHAR(64) NOT NULL REFERENCES inspections(id) ON DELETE CASCADE,
    type                VARCHAR(32) NOT NULL,
    length_cm           DOUBLE PRECISION,
    width_mm            DOUBLE PRECISION,
    severity_class      VARCHAR(16) NOT NULL CHECK(severity_class IN ('hairline', 'moderate', 'severe')),
    confidence_score    DOUBLE PRECISION NOT NULL
);
```

4. Key Performance & Reliability Optimizations

To ensure rapid pipeline execution and database safety, the system implements the following structural refinements:

4.1 Quantized Texture Calculations

Converts standard grayscale patch crops down to 32 levels (gray // 8), reducing Gray-Level Co-occurrence Matrix (GLCM) cells from 65,536 to 1,024. This speeds up Haralick feature calculations by over 50x while preserving spatial contrast information.

4.2 SIFT Cache SHA-256 Keying

Implements a SIFTCache singleton in src/pipeline.py that caches keypoint descriptors keyed on SHA-256 file hashes rather than fragile filenames, preventing redundant SIFT computations on duplicate or identical frames.

4.3 Write-Ahead Logging (WAL) Mode

Configures WAL journal mode and SQLite event connects (PRAGMA journal_mode=WAL, PRAGMA synchronous=NORMAL) inside src/utils/sqlite_store.py to prevent database locked exceptions during concurrent writes under local testing.

4.4 Model Singleton Warmup

Loads ML model weights (ResNet-50 features & XGBoost classifier) as singletons at server startup to reduce per-patch classification latency from 4.0 s down to <50 ms.

4.5 Memory-First Pipelines

Uses io.BytesIO streams to build dynamic ReportLab PDF manuals, bypassing slow local disk writes.

5. Project Implementation & Integration Update

All integration issues between the frontend and backend have been resolved, and local dockerized services are running smoothly:

- **E2E Integration Verified**: The FastAPI API is fully integrated with Celery asynchronous tasks, Redis caching, and PostgreSQL storage. Upstream requests correctly route through Nginx.
- **Seeded Admin Credentials**: The default organization (cortex), user (admin@cortex.com / CortexPass123!), and building (Cortex Headquarters) are seeded. The Next.js frontend automatically performs a background login handshake to secure JWT authentication.
- **Dynamic 3D Models**: The CortexCrackAnalyzer.jsx component uses real-time metrics parsed from uploaded files to construct custom 3D model sandboxes depending on the detected defect's structure (Column, Slab, Beam).
- **Celery Scheduler Fixed**: Replaced the Celery Beat Django scheduler dependency with the native PersistentScheduler and bound it to write schedule logs to the writable /tmp/celerybeat-schedule folder, solving container write-permission blockages.
- **Production Build Compiled**: The Next.js static asset build has been exported to frontend/out and is actively served by Nginx.

6. Project Architecture Artifacts

For further high-density specifications and implementation guides, refer to the following local repository files:

- **system_architecture_design.md**: Deep-dive Kubernetes deployment configurations, database schemas, and caching layouts.
- **system_design_hld.md**: High-level monolithic processing pipeline and CV latency reduction explanations.
- **walkthrough.md**: Verification steps and developer health checklists.